Taylor Polynomials and Remainder Bounds
Answer Key

AP Calculus BC — Convergence and Taylor Series

Quick Answers

1. 3.25
2. $\frac{65}{24}$
3. $\frac{1841}{3840}$
4. $\frac{299}{384}$
5. $\frac{1}{192}$
6. $\frac{1}{2}$
7. $\frac{4}{703125}$
8. $\frac{1367}{7500}$
9. $\frac{1}{96}$
10. $\frac{143}{1500}$, $\frac{1}{40000}$

Common wrong answers

1. *If they got 5: used the raw derivative values as coefficients and never divided by the factorials*
5. *If they got 0.0416667: used n rather than n+1 in the Lagrange bound, so the power and the factorial were both one too small*
6. *If they got 0: kept only the linear term of the series for e^x, so the numerator collapsed to zero and the limit was reported as zero*
7. *If they got 0.00106667: bounded the error with the last term that was included instead of the first term that was left out*
9. *If they got 0.104167: used the polynomial degree n in place of n+1, dividing by 4! and raising to the fourth power*

1. T_3 about $x = 2$ from a table of derivatives.

The centre is $c = 2$, so every term carries $(x-2)^k$, and the kth derivative value is divided by $k!$:

$$T_3(x) = f(2) + f'(2)(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3$$
$$= 3 - (x-2) + \frac{4}{2}(x-2)^2 + \frac{12}{6}(x-2)^3 = 3 - (x-2) + 2(x-2)^2 + 2(x-2)^3$$

At $x = 2.5$ the increment is $x - 2 = 0.5$:

$$T_3(2.5) = 3 - 0.5 + 2(0.25) + 2(0.125) = 3 - 0.5 + 0.5 + 0.25$$

$\boxed{3.25}$

Common wrong answer 5: the derivative values were used as the coefficients directly, giving $3 - 0.5 + 4(0.25) + 12(0.125)$. The $k!$ is not decoration — it is exactly what makes $T_3^{(k)}(2)$ equal $f^{(k)}(2)$.

2. Fourth-degree Maclaurin polynomial for e^x at $x = 1$.

Every derivative of e^x is e^x and $e^0 = 1$, so every coefficient is $1/k!$:

$$T_4(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$$

At $x = 1$ each power is 1, so the value is the partial sum $\displaystyle\sum_{n=0}^{4} \frac{1}{n!}$:

$$1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} = \frac{24 + 24 + 12 + 4 + 1}{24}$$

$$\boxed{\frac{65}{24}}$$

$\dfrac{65}{24} = 2.708\overline{3}$ against $e = 2.71828\ldots$, so four terms are already accurate to about 0.01.

3. $\sin\left(\frac{1}{2}\right)$ from the degree-5 polynomial.

Only odd powers appear, so the fifth-degree polynomial has three terms:

$$P_5(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

Substituting $x = \frac{1}{2}$:

$$\frac{1}{2} - \frac{1}{8 \cdot 6} + \frac{1}{32 \cdot 120} = \frac{1}{2} - \frac{1}{48} + \frac{1}{3840}$$

Over the common denominator 3840:

$$\frac{1920 - 80 + 1}{3840}$$

$$\boxed{\frac{1841}{3840}}$$

Grading note: P_5 and P_6 are the same polynomial here, because the degree-6 coefficient of $\sin x$ is 0. A student who writes four terms has usually mis-indexed $(-1)^n$.

4. Degree-6 Maclaurin polynomial for e^{-x^2}.

Do not differentiate — substitute $u = -x^2$ into $e^u = 1 + u + \dfrac{u^2}{2!} + \dfrac{u^3}{3!} + \cdots$. Each u^n contributes x^{2n}, so terms through $n = 3$ give every power up to x^6:

$$P_6(x) = 1 - x^2 + \frac{x^4}{2} - \frac{x^6}{6}$$

At $x = \frac{1}{2}$:

$$1 - \frac{1}{4} + \frac{1}{16 \cdot 2} - \frac{1}{64 \cdot 6} = 1 - \frac{1}{4} + \frac{1}{32} - \frac{1}{384} = \frac{384 - 96 + 12 - 1}{384}$$

$$\boxed{\frac{299}{384}}$$

Why substitution and not differentiation: the fourth derivative of e^{-x^2} is already a four-term product-rule expression, and the sixth is worse. Composition with a known series is the whole reason this is a BC topic rather than an exercise in patience.

5. Lagrange bound for T_3 of e^x at $x = 0.5$.

(a) With $n = 3$ and $c = 0$, the remainder bound uses $n + 1 = 4$ in *both* places:

$$|R_3(0.5)| \le \frac{M\,|0.5 - 0|^4}{4!} = \frac{2 \cdot (0.5)^4}{24} = \frac{2 \cdot \frac{1}{16}}{24} = \frac{1}{8 \cdot 24}$$

$$\boxed{\frac{1}{192}}$$

(b) Model justification: every derivative of $f(x) = e^x$ is again e^x, so $f^{(4)}(z) = e^z$. On $[0, 0.5]$ the function e^z is positive and increasing, so its largest value on that interval is $e^{0.5} = 1.6487\ldots$, and $1.6487\ldots < 2$. Therefore $\left|f^{(4)}(z)\right| \le 2$ for every z in $[0, 0.5]$, which is exactly what M has to satisfy.

Common wrong answer $\frac{1}{24} \approx 0.0417$: the degree $n = 3$ was used in place of $n + 1$, giving $2(0.5)^3/3!$. Both the exponent and the factorial step up by one.

Manual review: part (b) is a written justification, read by a teacher; part (a) is machine-checked. Any correct argument for a legitimate M earns full credit — $M = e$ or $M = 1.65$ would also be valid.

6. A limit by series, not by L'Hôpital.

Replace e^x by its Maclaurin series and cancel against the terms subtracted from it:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$e^x - 1 - x = \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots$$

The leading behaviour of the numerator is $x^2/2$, matching the denominator's degree exactly. Divide term by term:

$$\frac{e^x - 1 - x}{x^2} = \frac{1}{2} + \frac{x}{6} + \frac{x^2}{24} + \cdots$$

Every remaining term carries a positive power of x and vanishes as $x \to 0$.

$\boxed{\frac{1}{2}}$

Common wrong answer 0: the series was truncated at $1 + x$, so the numerator appeared to be exactly 0. The whole point of the series method is that the *first surviving term* decides the limit, so you must keep terms until something fails to cancel.

7. Alternating series bound for $\cos\left(\frac{2}{5}\right)$.

$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$, so the fourth-degree polynomial stops after $\frac{x^4}{4!}$ and the *first omitted* term is $\frac{x^6}{6!}$.

The series alternates and, for $x = \frac{2}{5}$, its terms decrease in size, so the alternating series error bound applies:

$$|R_4| < \left|\frac{(2/5)^6}{6!}\right| = \frac{64/15625}{720} = \frac{64}{11{,}250{,}000}$$

$\boxed{\frac{4}{703125}}$

That bound is about 5.7×10^{-6}, so a degree-4 polynomial already pins $\cos(0.4)$ to five decimal places.

Common wrong answer ≈ 0.00107*:* the last *included* term $\frac{(2/5)^4}{4!}$ was used as the bound. The theorem bounds the error by the first term you threw away, never by the last one you kept.

8. $\ln(1.2)$ from four terms.

$\ln(1.2) = \ln(1 + 0.2)$, so substitute $x = \frac{1}{5}$ into the given series and stop after four terms:

$$\frac{1}{5} - \frac{(1/5)^2}{2} + \frac{(1/5)^3}{3} - \frac{(1/5)^4}{4} = \frac{1}{5} - \frac{1}{50} + \frac{1}{375} - \frac{1}{2500}$$

Over the common denominator 7500:

$$\frac{1500 - 150 + 20 - 3}{7500}$$

$$\boxed{\frac{1367}{7500}}$$

$\frac{1367}{7500} = 0.18226\overline{6}$ against $\ln(1.2) = 0.182321\ldots$ The next term, $\frac{(1/5)^5}{5} = 0.000064$, bounds that gap — and it does.

9. Lagrange bound with a supplied derivative bound.

Here $n = 4$, $c = 1$, $x = 1.5$, and $M = 40$ is handed to you, so the only work is putting $n + 1 = 5$ into both the exponent and the factorial:

$$|R_4(1.5)| \leq \frac{40\,|1.5 - 1|^5}{5!} = \frac{40 \cdot (0.5)^5}{120} = \frac{40 \cdot \frac{1}{32}}{120} = \frac{1.25}{120}$$

$$\boxed{\frac{1}{96}}$$

Common wrong answer ≈ 0.104*:* the exponent and factorial were left at $n = 4$, giving $40(0.5)^4/4!$ — ten times too large. Reading $|R_4|$ as "use 4" is the single most common Lagrange slip.

Note that M bounds the $(n+1)$st derivative, here $g^{(5)}$, and that the bound is required on the whole interval between c and x, not just at the endpoints.

10. Synthesis — $\ln(1.1)$ with T_3, its error bound, and its direction.

(a) From $\ln(1+x) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3} - \cdots$, the third-degree Maclaurin polynomial is

$$T_3(x) = x - \frac{x^2}{2} + \frac{x^3}{3}$$

$\ln(1.1) = \ln(1+0.1)$, so put $x = \frac{1}{10}$:

$$\frac{1}{10} - \frac{1}{200} + \frac{1}{3000} = \frac{300 - 15 + 1}{3000} = \frac{286}{3000}$$

$$\boxed{\frac{143}{1500}}$$

(b) The series alternates and its terms decrease in size, so the error is less than the first omitted term, $\dfrac{x^4}{4}$ at $x = \frac{1}{10}$:

$$|R_3| < \frac{(1/10)^4}{4} = \frac{1}{10{,}000 \cdot 4}$$

$$\boxed{\text{error} < \frac{1}{40000}}$$

(c) Model answer: an **overestimate**. The first omitted term is $-\dfrac{x^4}{4}$, which is *negative* at $x = \frac{1}{10}$, and for an alternating series with terms decreasing in size the true value always lies between consecutive partial sums — so the true value sits below the partial sum that stops at $+\dfrac{x^3}{3}$. Therefore $\ln(1.1) < \dfrac{143}{1500}$.

Confirmation for the grader only: $\ln(1.1) = 0.0953102\ldots$ and $\dfrac{143}{1500} = 0.0953\overline{3}$, a gap of about 2.3×10^{-5} — below the bound from part (b), and on the high side, as predicted. Students were told not to use this route, and an answer resting only on a calculator value should not receive full credit for (c).

Manual review: part (c) is written reasoning about the sign of the remainder, read by a teacher; parts (a) and (b) are machine-checked.